Decision tree vs Random forest

Introduction:

Our goal is to classify breast cancer dataset which is provided by SKlearn

This dataset consists from:

```
Train set size: 398 samples
Validation set size: 85 samples
Test set size: 86 samples
Number of features: 30
```

We need to classify whether this cancer is benign or malignant

As we can see our dataset is relatively small

The best classifiers we can use for this task is random forest or decision tree

Decision Trees:

It mainly depends on entropy to find the maximum information gain by searching

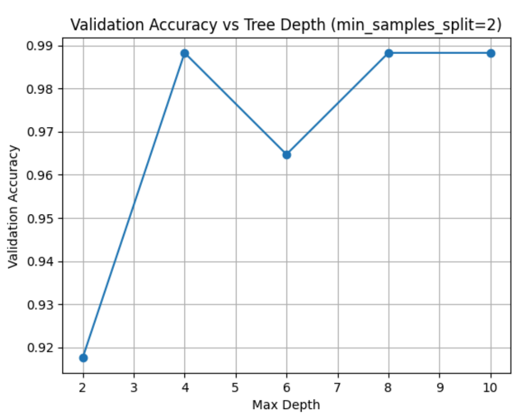
The best features and best threshold and the splitting on them

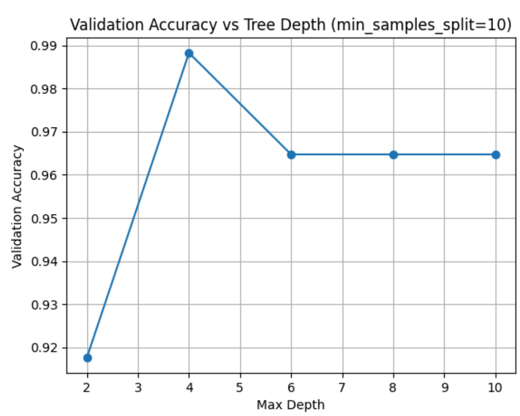
The model provided:

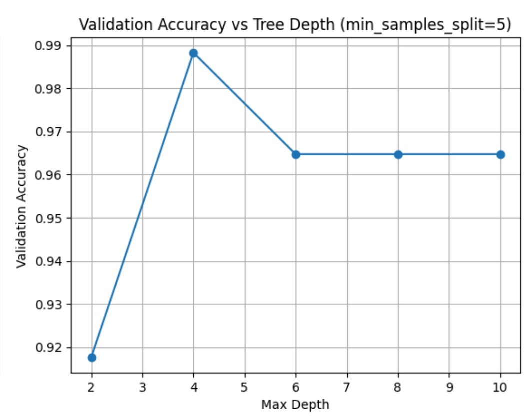
```
Training Accuracy: 0.9522613065326633
Validation Accuracy(without tuning): 0.9176470588235294
Best Validation Accuracy: 0.9882352941176471
max_depth: 4, min_samples_split: 2

Test Accuracy: 0.8837209302325582
```

Accuracy vs depth:







Random Forest:

Forest depends on generating many different trees with different number of training samples to generalize more and don't overfit and after training those Trees they makes a votes and the majority of the vote we will classify to this class

This model provided:

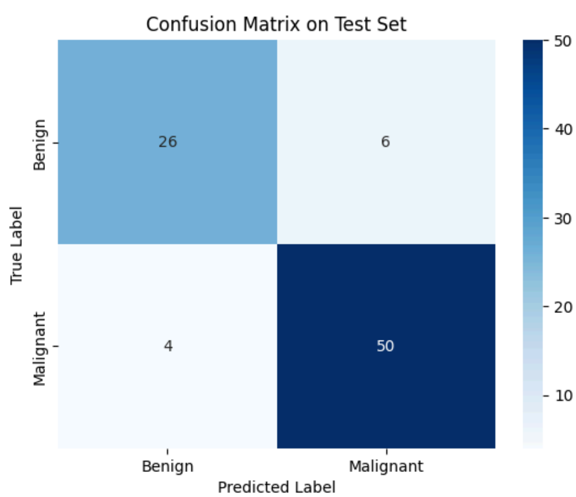
```
Best Validation Accuracy: 0.9882352941176471
Best Random Forest Params (T, max_features): (30, 5)
Final Test Accuracy: 0.8953488372093024
Accuracy: 0.8953488372093024
Precision (Benign, Malignant): [0.89655172 0.89473684]
Recall (Benign, Malignant): [0.8125     0.94444444]
F1-score (Benign, Malignant): [0.85245902 0.91891892]
```

Bias and Variance:

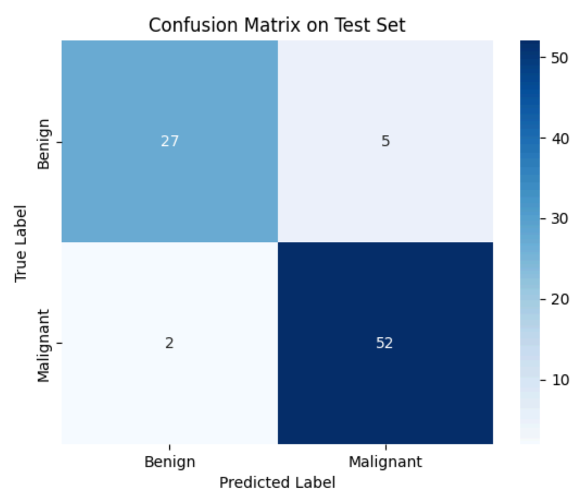
Bias is the error from simplifying the true relationship in the data,resulting in a model that consistently misses the mark (**underfitting**).

Variance is the error from the model being too sensitive to small fluctuations in the training data, resulting in wildly different predictions for the same input (**overfitting**).

Comparison:



Constrained Decision Tree



Random Forest

Feature	Unconstrained Decision Tree	Constrained Decision Tree	Random Forest (Ensemble)
Model Goal	Maximize fit to training data.	Balance fit and generalization	Minimize total error (especially variance).
Complexity	Very High (Grows until leaves are pure or minimal).	Medium (Limited by max_depth or min_samples_split).	High (Ensemble of Medium/High-Bias Trees).
Bias	Very Low. Can fit complex boundaries perfectly.	Low / Moderate. Slightly higher than unconstrained, due to simplification.	Low / Moderate. The bias of the individual base trees is slightly increased due to constraints and sub-sampling.
Variance	Very High. Extremely sensitive to training data noise (memorization).	Lower. Pruning/constraints stabilize the tree's structure.	Very Low. Variance of individual trees is averaged out through majority voting.
Generalization	Poor (High risk of Overfitting).	Good (Aims for the optimal balance).	Excellent (Highest overall test accuracy).
Techniques Used	No regularization.	Pruning (max_depth,min_samples_split).	Bagging (Bootstrap Sampling) & Feature Randomness + Majority Voting.